SalesGenie AI

AI Sales Assistant & Lead Intelligence Platform

Intern Training Guide

Python · FastAPI | OpenAI LLM | Streamlit | PostgreSQL

Milestone-based · 8 Weeks · Full Stack AI Project

OVERVIEW

What is SalesGenie AI?

The Problem

- Sales teams waste hours researching prospects manually
- Inconsistent outreach leads to low conversion rates
- No central system to track leads + AI insights together
- Follow-ups are forgotten or poorly timed

✦ SalesGenie AI Solves This

🔍 AI analyses every prospect company

📊 Scores leads 0-100 automatically

✉️ Generates personalised cold emails

💬 Summarises sales calls & meetings

📈 Live analytics dashboard

MODULES

6 Core Modules

L

Lead Management

Create, search and track all your prospects in a central database

I

Lead Intelligence

AI analyses company profiles and generates business insights

E

AI Outreach

Personalised cold emails generated from prospect data

S

Lead Scoring

Predict conversion probability (0-100) with AI scoring engine

C

Conversation Intel

Summarise sales calls and extract action items automatically

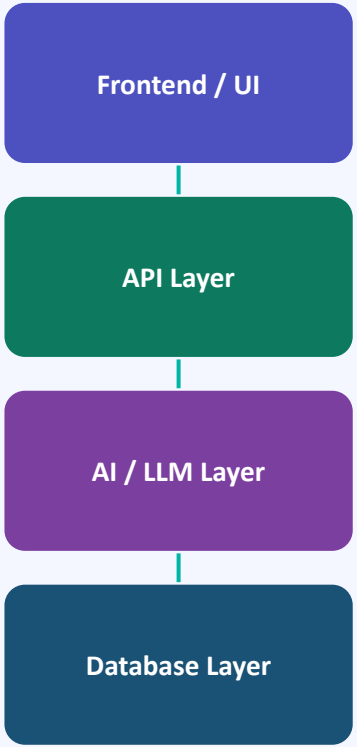
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Analytics Dashboard

Streamlit dashboard with pipeline, scores and performance KPIs

TECH STACK

Technology Stack



- Streamlit (Python dashboard)
- Plotly charts
- Pandas data tables

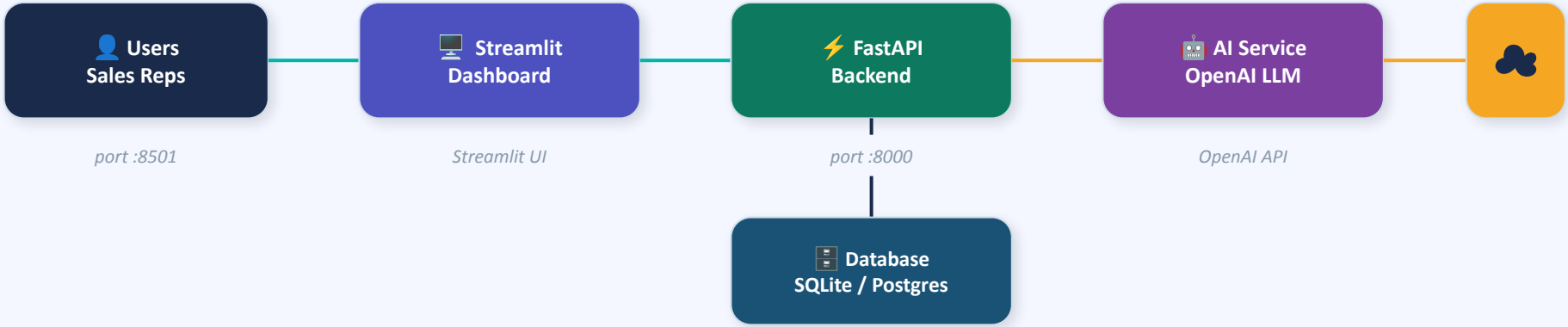
- FastAPI (Python)
- Pydantic validation
- CORS middleware

- OpenAI GPT-3.5 / GPT-4
- LangChain agents
- Prompt engineering

- SQLAlchemy ORM
- SQLite (dev) / PostgreSQL (prod)
- Alembic migrations

ARCHITECTURE

System Architecture



Project File Layout

`main.py`	FastAPI entry point — registers routers, creates DB tables
`app/database.py`	DB connection — SQLite default, Postgres via .env
`app/models/models.py`	8 database tables (Lead, LeadScore, CompanyInsight…)
`app/services/ai_service.py`	All OpenAI calls — scoring, outreach, summarization
`app/api/leads.py`	CRUD endpoints: POST/GET/PUT/DELETE /leads
`app/api/intelligence.py`	AI endpoints: analyze, score, outreach, summarize
`dashboard/app.py`	Streamlit UI — 5 pages, connects to port 8000

INSTALLATION

Step-by-Step Installation

1

Prerequisites — install these first

```
python --version    # must be 3.10+
node --version      # must be 18+
git --version       # any version
```

2

Clone & navigate into the project

```
git clone https://github.com/your-repo/salesgenie-ai.git
cd salesgenie-ai
```

3

Create & activate virtual environment

```
python -m venv venv
source venv/bin/activate       # Mac / Linux
venv\Scripts\activate          # Windows
```

4

Install all Python packages

```
pip install -r requirements.txt
```

5

Configure environment & seed data

```
cp .env.example .env      # then add your OPENAI_API_KEY
python seed_data.py          # loads 5 sample leads
```

RUNNING THE APP

Two Terminals, Two Servers

Terminal 1 — Backend

```
$ cd salesgenie-ai
$ source venv/bin/activate
(venv) $ uvicorn main:app \
         --reload --port 8000

INFO: Application startup complete.
INFO: Uvicorn running on
      http://127.0.0.1:8000

→ Open: localhost:8000/docs
```

Terminal 2 — Dashboard

```
$ cd salesgenie-ai
$ source venv/bin/activate
(venv) $ streamlit run \
         dashboard/app.py

You can now view your Streamlit app
in your browser.
  Local URL: localhost:8501

→ Dashboard opens automatically!
```

💡 No OpenAI key? The app runs in MOCK MODE — AI returns placeholder text. Add OPENAI_API_KEY to .env for real AI responses.

DATABASE

Database Schema

leads

- 🔑 **lead_id PK**
- company_name
- industry
- contact_name
- lead_status
- tech_stack
- funding_stage

lead_scores

- 🔑 **score_id PK**
- 🔗 lead_id FK
- lead_score 0-100
- conversion_probability
- priority_level

company_insights

- 🔑 **insight_id PK**
- 🔗 lead_id FK
- business_needs
- opportunities
- pain_points

outreach_campaigns

- 🔑 **campaign_id PK**
- 🔗 lead_id FK
- email_subject
- email_content
- follow_up_strategy

sales_interactions

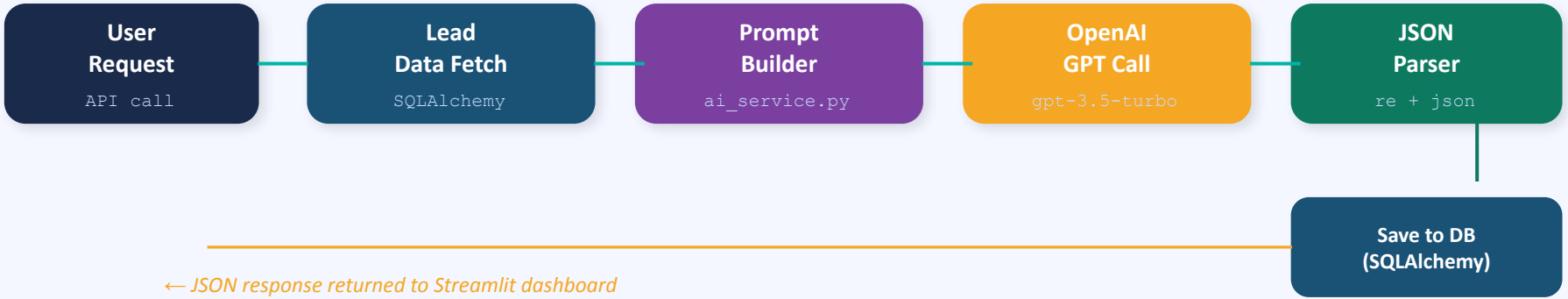
- 🔑 **interaction_id PK**
- 🔗 lead_id FK
- summary
- action_items
- key_points

sales_analytics

- 🔑 **analytics_id PK**
- 🔗 user_id FK
- conversion_rate
- pipeline_value

HOW THE AI WORKS

AI Agent Flow



Code: How ai_service.py calls OpenAI

```
response = client.chat.completions.create(
    model="gpt-3.5-turbo",
    messages=[{"role":"system","content":system}, {"role":"user","content":user}],
    max_tokens=600
)
```

AI AGENT DEEP DIVE

Lead Scoring Agent

How the Scoring Prompt Works

SYSTEM PROMPT

```
"You are an AI lead qualification engine.
Score the lead from 0-100 and estimate
conversion probability (0.0-1.0).
Return ONLY a JSON object with keys:
lead_score, conversion_probability,
priority_level, scoring_factors."
```

USER PROMPT (filled with lead data)

```
"Company: TechCorp Solutions
Industry: Enterprise Software
Size: 250-500 employees
Revenue: $45M-$60M
Funding: Series C
Tech Stack: AWS, Python, React
Status: new"
```

AI JSON Response

```
{
  "lead_score": 87,
  "conversion_probability": 0.78,
  "priority_level": "high",
  "scoring_factors": {
    "company_growth": "+25 pts",
    "funding_stage": "+20 pts"
  }
}
```

Key Scoring Factors

+20	**Funding Stage** Series C/D = high budget	+15	**Company Size** 250-500 = decision makers
+18	**Revenue** $45M+ = real budget	+12	**Tech Stack Match** Our tools compatible
+10	**Lead Status** Active engagement	+12	**Industry Fit** Known buyer personas

API REFERENCE

FastAPI Endpoints — curl Examples

GET `/health`

Health check

```
curl http://localhost:8000/health
```

POST `/leads/`

Add a lead

```
curl -X POST http://localhost:8000/leads/ \
  -H "Content-Type: application/json" \
  -d '{"company_name":"AcmeCorp","industry":"SaaS"}'
```

GET `/leads/`

List all leads

```
curl http://localhost:8000/leads/
```

POST `/intelligence/score/{id}`

AI score a lead

```
curl -X POST http://localhost:8000/intelligence/score/1
```

POST `/intelligence/analyze/{id}`

Company analysis

```
curl -X POST http://localhost:8000/intelligence/analyze/1
```

POST `/intelligence/outreach`

Generate email

```
curl -X POST http://localhost:8000/intelligence/outreach \
  -H "Content-Type: application/json" \
  -d '{"lead_id":1,"tone":"professional"}'
```

💡 Visit localhost:8000/docs for the interactive Swagger UI — test every endpoint in the browser!

AI AGENTS

Outreach & Conversation Agents

Email Outreach Agent

Select Lead → Fetch Insight → Build Prompt → GPT Call → Save Campaign

lead_id from DB | CompanyInsight | tone + focus | gpt-3.5-turbo | OutreachCampaign

Generated Email Preview

Subject: Transform Your Data Pipeline with AI

Hi Sarah,
Congrats on closing Series C! As you scale, efficient data handling becomes critical — that's exactly where we help.
Similar companies have cut data processing time by 65% using our platform.
[AI-personalised from company profile]

Conversation Intelligence Agent

Paste Transcript → AI Summarize → Extract Items → Save + Return

raw call notes | GPT prompt | JSON parse | SalesInteraction

AI-Generated Meeting Summary

Summary:

Sarah is evaluating our platform for data analytics. Budget approved for Q3. Competitor evaluation in progress.

✅ **Action Items:**

- Send technical architecture doc by Jun 10
- Schedule deep-dive with engineering team

```
# app/api/intelligence.py — Outreach endpoint (simplified)
@router.post("/outreach")
def generate_outreach(payload: OutreachRequest, db: Session = Depends(get_db)):
    lead = db.query(Lead).filter(Lead.lead_id == payload.lead_id).first()
    result = ai_service.generate_outreach_email(lead_dict, insight_dict, tone=payload.tone)
    campaign = OutreachCampaign(lead_id=payload.lead_id, **result)
    db.add(campaign);  db.commit()
    return campaign
```

FRONTEND

Streamlit Dashboard – 5 Pages



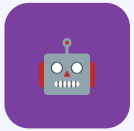
Page 1: Dashboard

Pipeline overview, conversion rate, hot/warm/cold lead split, top-10 leads ranked by AI score



Page 2: Leads

Search/filter all prospects. Add new leads via form. View full lead records with all company details



Page 3: AI Tools

Run company analysis & lead scoring per prospect. Gauge chart shows score. Next-best-action recommendation



Page 4: Outreach

Select a lead, choose tone (professional/friendly/urgent) and generate personalised cold email instantly



Page 5: Conversations

Paste a call transcript, click Summarize — get meeting summary, action items and key points in seconds

`dashboard/app.py — each page is an `elif page ==` branch. All data comes from FastAPI via requests.get() / requests.post()`

MILESTONES

8-Week Build Plan

M1 · Weeks 1–2

Lead Management & Intelligence Engine

- Build Lead & User database tables
- FastAPI CRUD: POST/GET/PUT/DELETE /leads
- Company profile analysis with LLM
- Generate lead intelligence insights

✓ ≥95% of prospect records managed and analysed correctly

M2 · Weeks 3–4

Outreach Generation & Lead Scoring

- AI lead scoring engine (0-100 scale)
- Conversion probability prediction
- Personalised cold email generator
- Follow-up strategy recommendations

✓ Lead scoring accuracy ≥85% · Outreach content for multiple industries

M3 · Weeks 5–6

CRM Integration & Conversation Intel

- Sales call/meeting summarisation
- Action item extraction from transcripts
- FastAPI webhook endpoints for CRM
- CRMSyncLog table + HubSpot-ready structure

✓ CRM sync functional · Meeting summaries generated correctly

M4 · Weeks 7–8

Dashboard & Final Deployment

- Streamlit 5-page analytics dashboard
- Automated follow-up recommendation engine
- End-to-end testing (API + UI)
- Deployment on Railway / Render / AWS

✓ Dashboard fully operational · Lead analysis <5 seconds · User satisfaction ≥85%

Start Building Today

Your 4 terminal commands to get started right now

1. `$ python -m venv venv && source venv/bin/activate`
2. `$ pip install -r requirements.txt`
3. `$ python seed_data.py`
4. `$ uvicorn main:app --reload --port 8000`

API Docs: localhost:8000/docs

Dashboard: localhost:8501

Swagger: localhost:8000/redoc